

Session 4: Data Visualization with matplotlib and seaborn

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Data Visualization

Data visualization helps in understanding patterns, trends, and relationships in data.

It is a crucial element in scientific research, enabling researchers to interpret and communicate their results effectively

Types of Data Visualization

1. Univariate Data Visualizations (Single Variable)

- ✓ **Histogram:** Used for understanding the distribution of a single variable.
- ✓ **Box Plot:** Used for Detecting outliers and understanding the spread of data.

2. Bivariate Data Visualizations (Two Variables)

- ✓ **Scatter Plot:** Used for understanding relationships between two numerical variables.
- ✓ **Line Plot:** Used for showing trends over time or continuous data.
- ✓ **Bar Chart:** Used for comparing categorical data.

3. Multivariate Data Visualizations (More than Two Variables)

- ✓ **Heatmap:** Used for visualizing correlations between multiple numerical variables.
- ✓ **Pair Plot:** Used for visualizing pairwise relationships in the dataset.
- ✓ **Violin Plot:** Used for understanding the distribution of a variable across categories.

4. Specialized Data Visualizations

- ✓ **Pie Chart:** Used for representing proportions.
- ✓ **Bubble Chart:** Used for adding a third variable to a scatter plot(Comparing three numerical variables)

✔ **Word Cloud:** used to highlight keywords, trends, or themes in textual data (Text Analysis : Highlighting key terms in articles, reviews, or social media posts.)

5. Time Series Visualizations

✔ **Time Series Line plot:** Used to analyze trends, patterns, or changes in data over a continuous period (e.g., days, months, years).

✔ **Autocorrelation Plot:** Used for finding patterns in time series data.

Best Practices

✔ Choose the Right Chart Type (e.g., bar charts for categories, line charts for trends).

✔ Follow Design Principles (simplicity, consistency, accessibility).

✔ Use Storytelling to highlight key insights and structure visuals logically.

✔ Avoid Common Pitfalls (misleading scales, cluttered visuals, unnecessary 3D charts).

Import necessary Libraries

```
In [1]: import pandas as pd          # Importing pandas for data manipulation
import numpy as np                # Importing NumPy for numerical computations
import seaborn as sns            # Importing seaborn for data visualization
import matplotlib.pyplot as plt  # Import the pyplot module
import os                        # provides functions for interacting with the op
import warnings
warnings.simplefilter(action='ignore')
```

```
In [2]: # Class boundaries and frequencies
class_boundaries = [299.5, 309.5, 319.5, 329.5, 339.5, 349.5, 359.5]
frequencies = [4, 6, 10, 1, 5, 4]

# Midpoints for frequency polygon
midpoints = [304.5, 314.5, 324.5, 334.5, 344.5, 354.5]
polygon_frequencies = [4, 6, 10, 1, 5, 4]

# Extra points to close the polygon
midpoints_extended = [294.5] + midpoints + [364.5]
frequencies_extended = [0] + polygon_frequencies + [0]

# Create the plot
plt.figure()

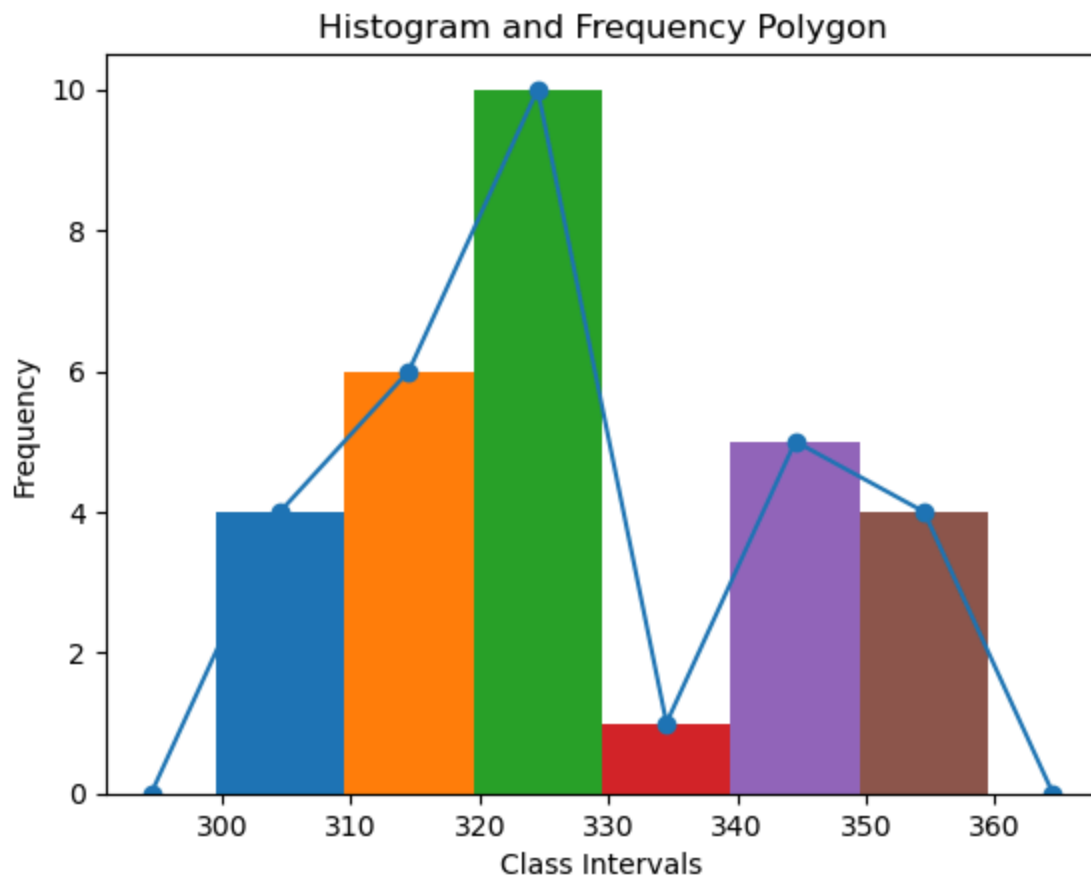
# Draw histogram bars using class boundaries
for i in range(len(frequencies)):
    plt.bar(class_boundaries[i], frequencies[i],
            width=10, align='edge')

# Draw frequency polygon
```

```
plt.plot(midpoints_extended, frequencies_extended, marker='o')

# Labels and title
plt.xlabel("Class Intervals")
plt.ylabel("Frequency")
plt.title("Histogram and Frequency Polygon")

plt.show()
```



Import dataset

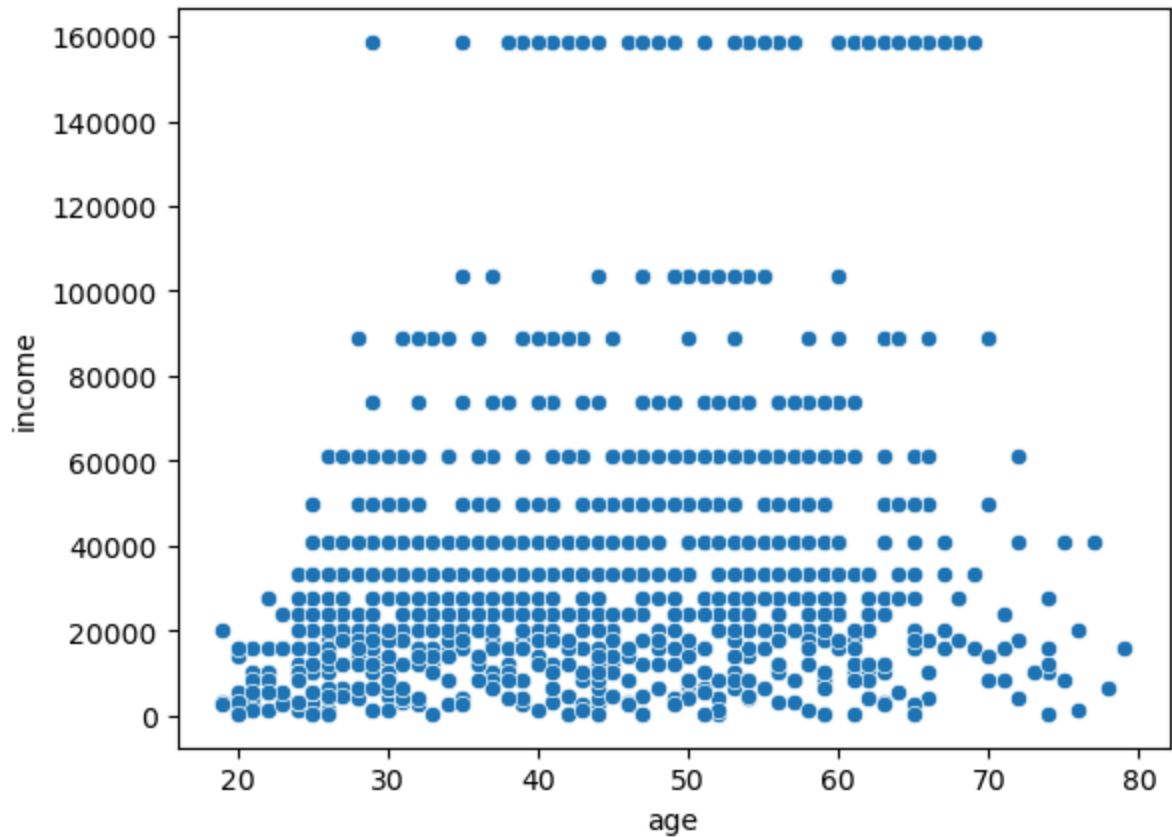
```
In [3]: gss = pd.read_csv("GSSsubset.csv")
gss.head()
```

```
Out[3]:
```

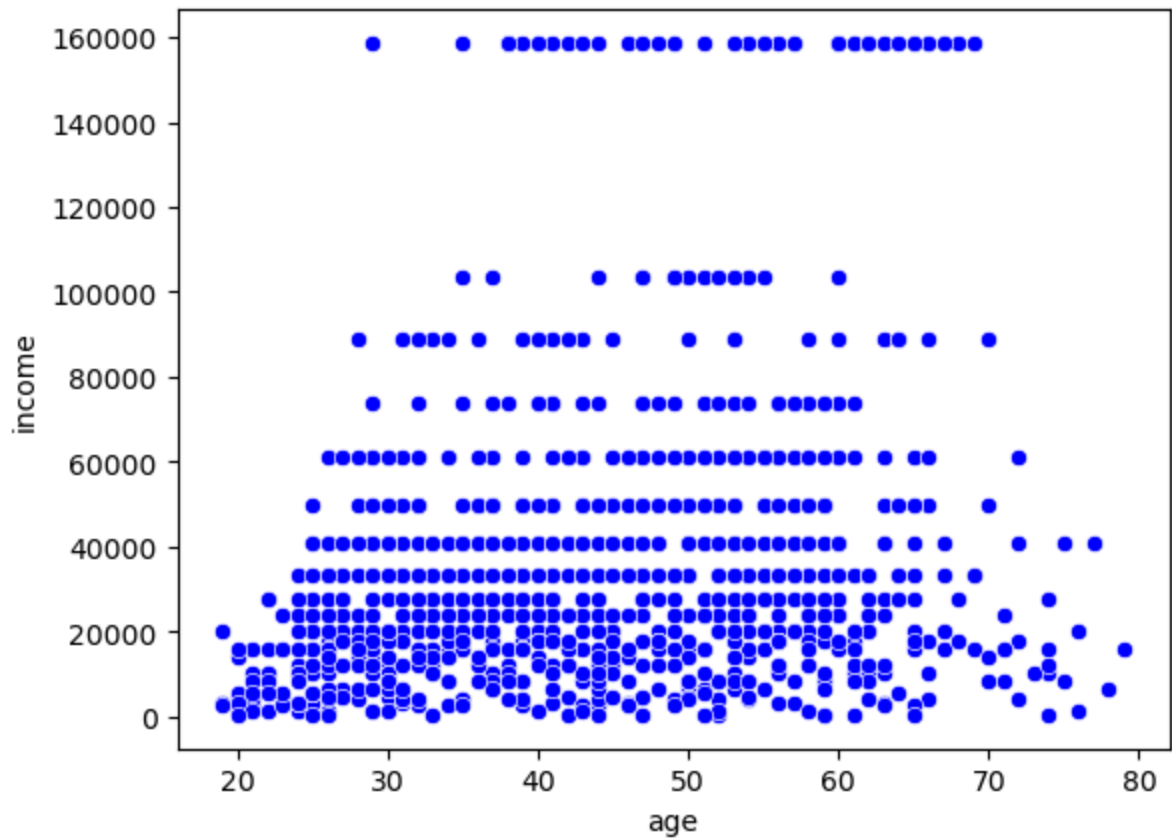
	id	sex	degree	income	marital	age	height	weight	hrswrk
0	1	MALE	BACHELOR	60967.50	DIVORCED	53	72	190	60
1	2	FEMALE	BACHELOR	60967.50	MARRIED	26	60	97	40
2	4	FEMALE	BACHELOR	10161.25	MARRIED	56	68	160	20
3	14	FEMALE	HIGH SCHOOL	17551.25	MARRIED	40	65	156	37
4	16	MALE	HIGH SCHOOL	17551.25	MARRIED	56	66	210	6

Scatter Plot: Income vs Age

```
In [4]: import seaborn as sns
sns.scatterplot(data = gss, x='age', y='income')
plt.show()
```



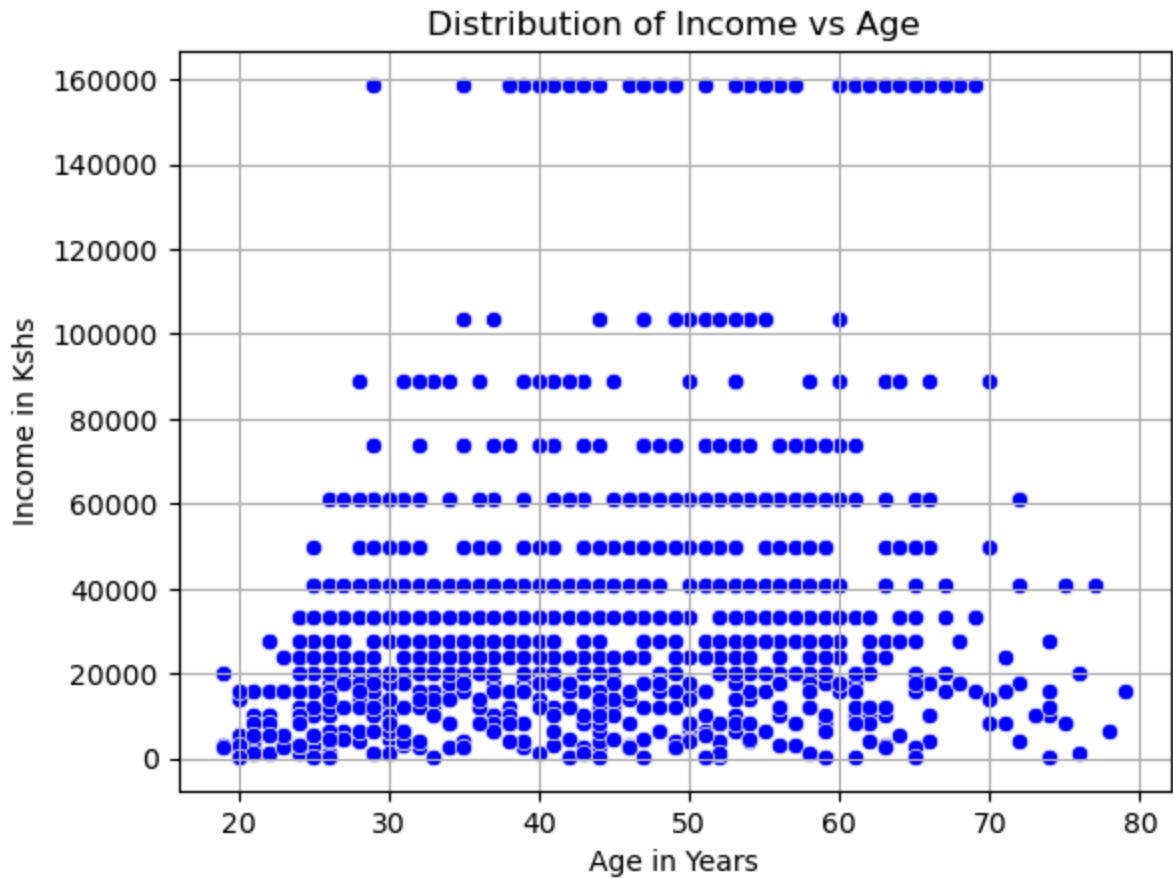
```
In [5]: sns.scatterplot(data=gss, x='age', y='income', color='blue')
plt.show()
```



In []:

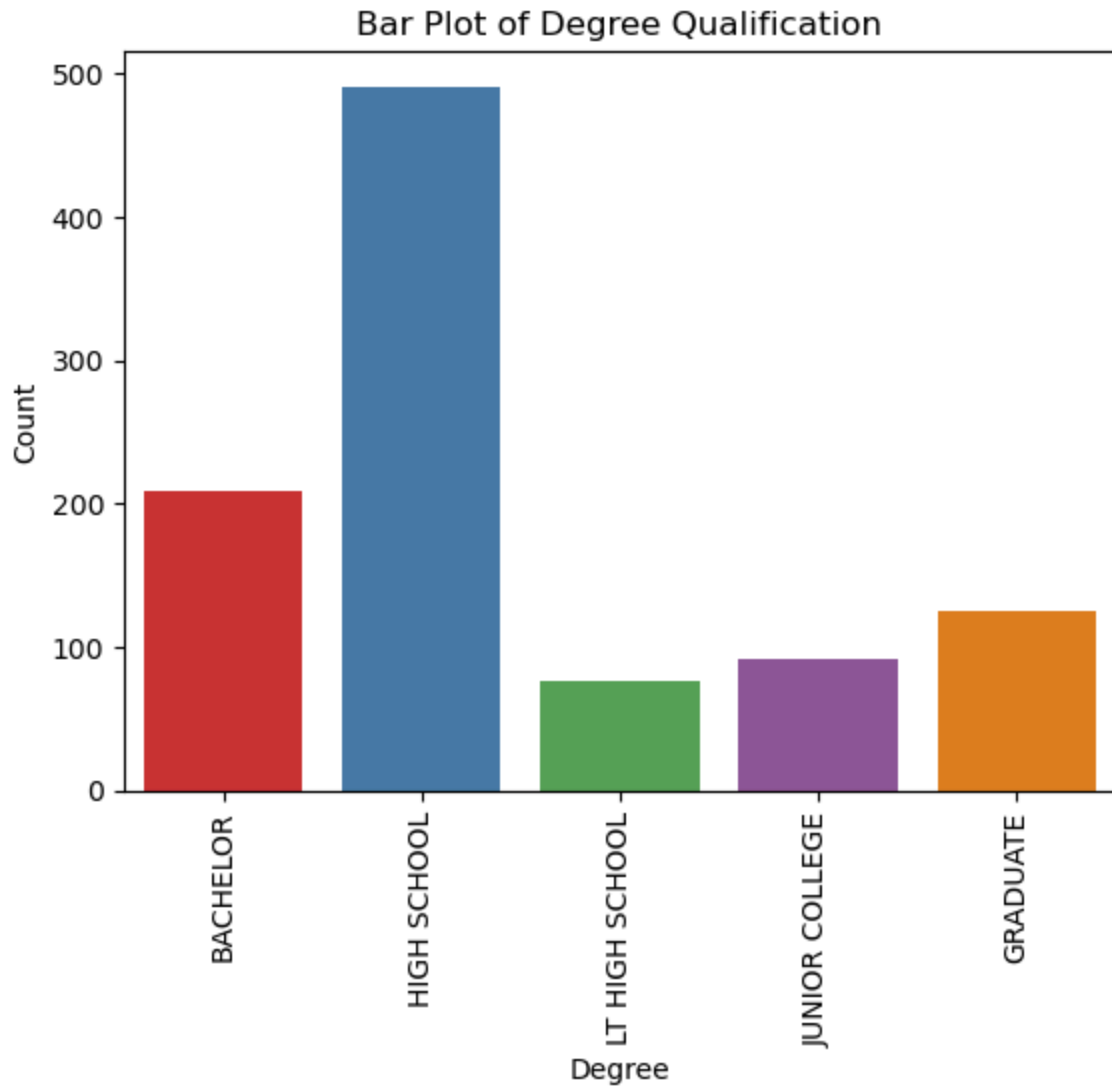
```
In [6]: sns.scatterplot(data=gss, x='age', y='income', color='blue')

plt.title("Distribution of Income vs Age")
plt.xlabel("Age in Years")
plt.ylabel("Income in Kshs")
plt.grid(True)
plt.show()
```



Bar Plot: Degree Qualification Count

```
In [7]: sns.countplot(data=gss, x='degree', palette='Set1')
plt.title("Bar Plot of Degree Qualification")
plt.xlabel("Degree")
plt.ylabel("Count")
plt.xticks(rotation=90)
plt.show()
```

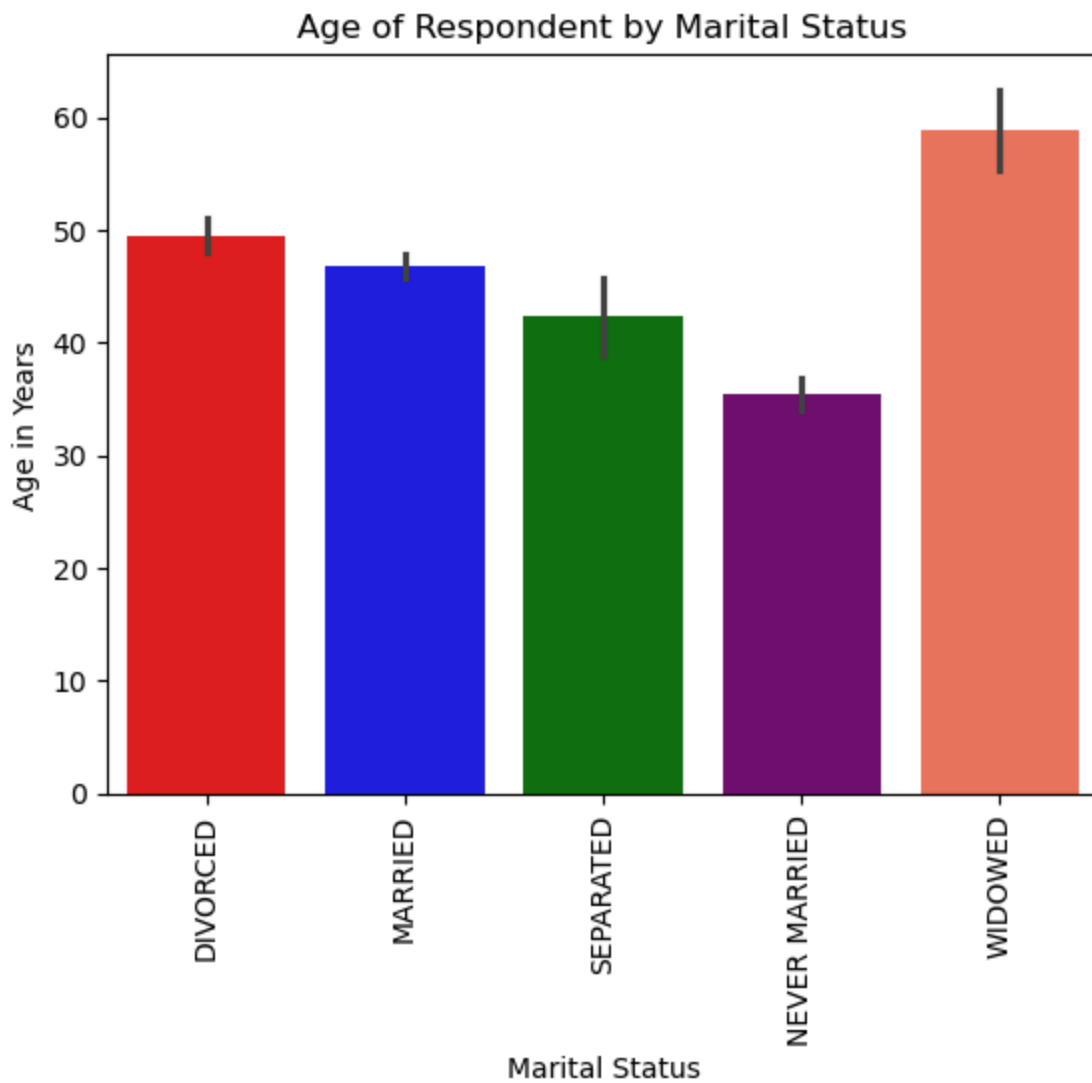


```
In [8]: # Create a bar chart on marital status
sns.barplot(data=gss, x='marital', y='age', hue='marital', palette=['red', 'blue',

# Add label and title for better readability
plt.xlabel('Marital Status')
plt.ylabel('Age in Years')
plt.title('Age of Respondent by Marital Status')

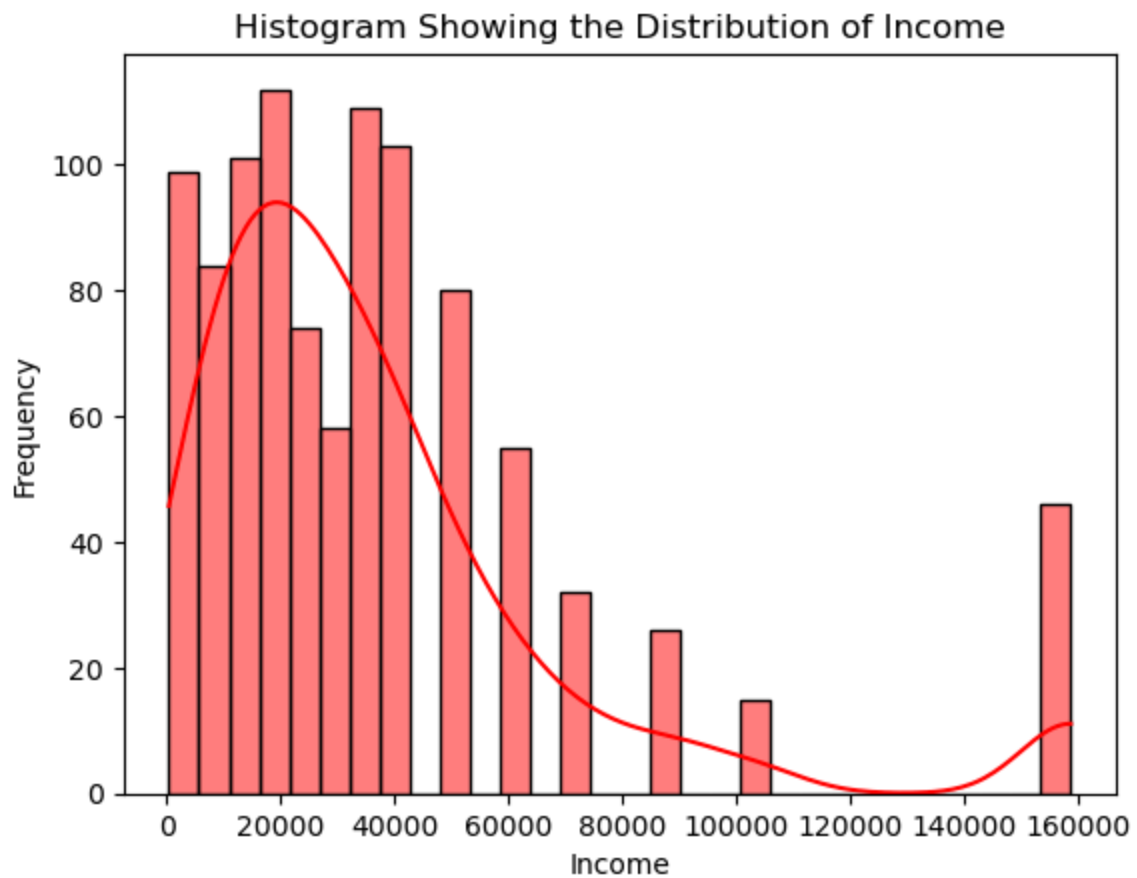
# Rotate x-axis labels if needed
plt.xticks(rotation=90)

# display the plot
plt.show()
```



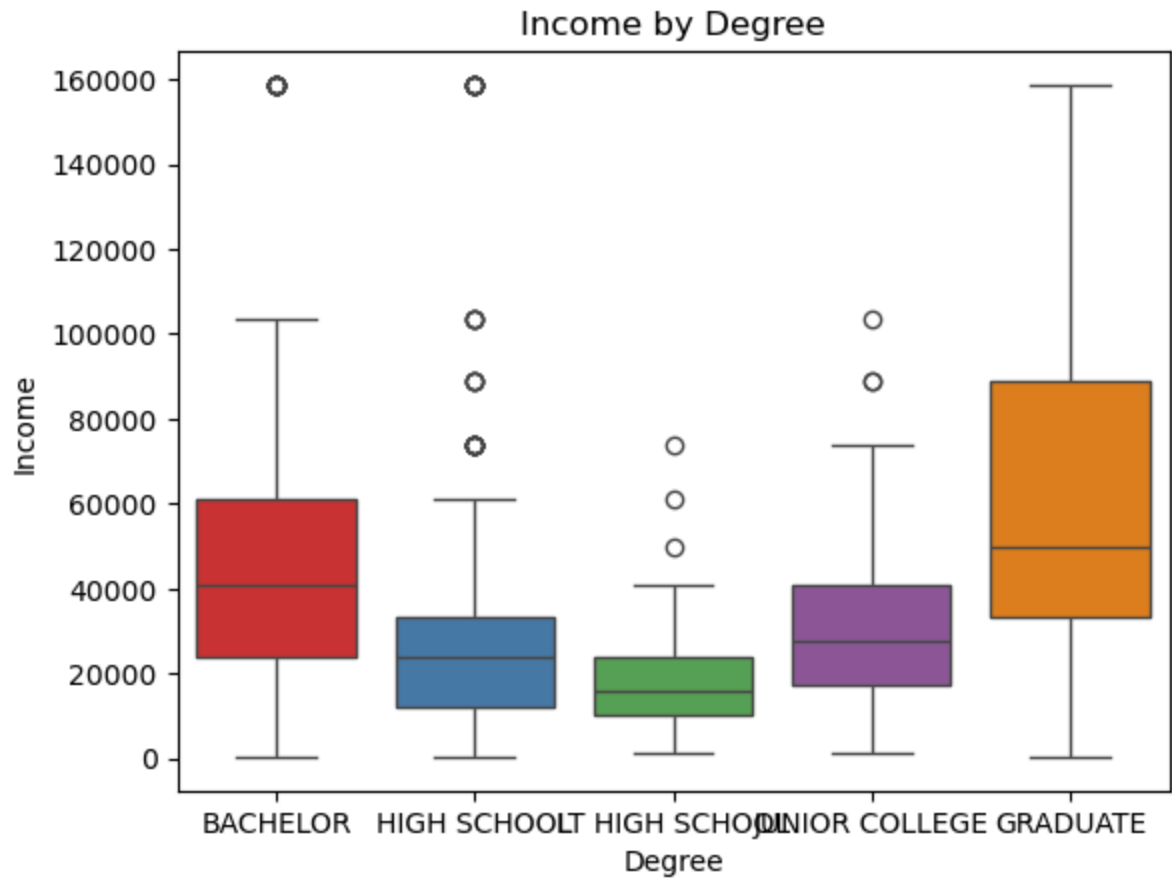
Histogram: Income Distribution

```
In [9]: sns.histplot(gss, x='income', bins=30, kde=True, color="red")
plt.title("Histogram Showing the Distribution of Income")
plt.xlabel("Income")
plt.ylabel("Frequency")
plt.show()
```

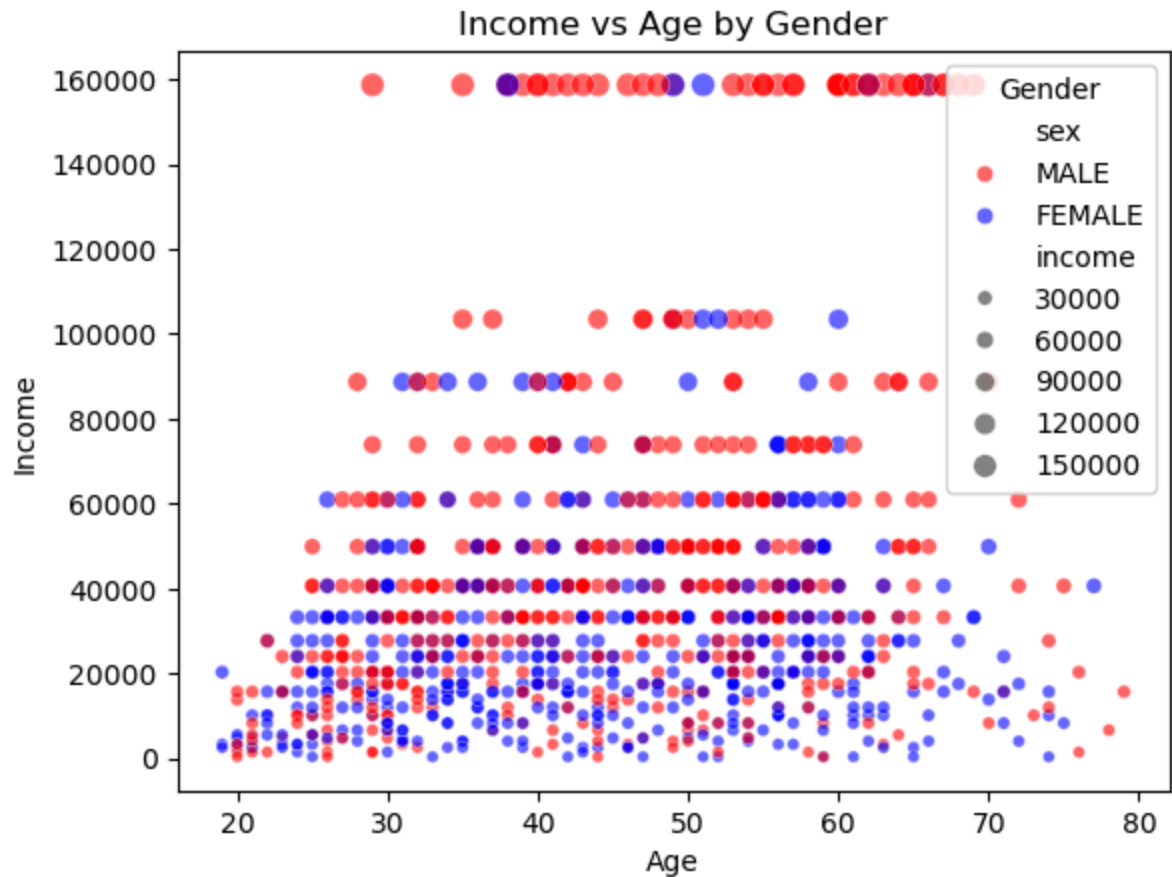
Boxplot: Income by Degree

```
In [10]: sns.boxplot(data=gss, x='degree', y='income', palette='Set1')
plt.title("Income by Degree")
plt.xlabel("Degree")
plt.ylabel("Income")
plt.show()
```



Customizing Scatter Plot with Color and Size

```
In [11]: sns.scatterplot(data=gss, x='age', y='income', hue='sex', size='income', alpha=0.6,
plt.title("Income vs Age by Gender")
plt.xlabel("Age")
plt.ylabel("Income")
plt.legend(title='Gender')
plt.show()
```



Homework: using German credit dataset

1. Create scatter plot for a different dataset
2. Create a bar plot for a categorical variable
3. Create a histogram for a single variable
4. Create a boxplot to compare distributions
5. Customize plots with titles, labels, themes, and colors

ADVANCED CHART

Scatter Plot with Regression Line

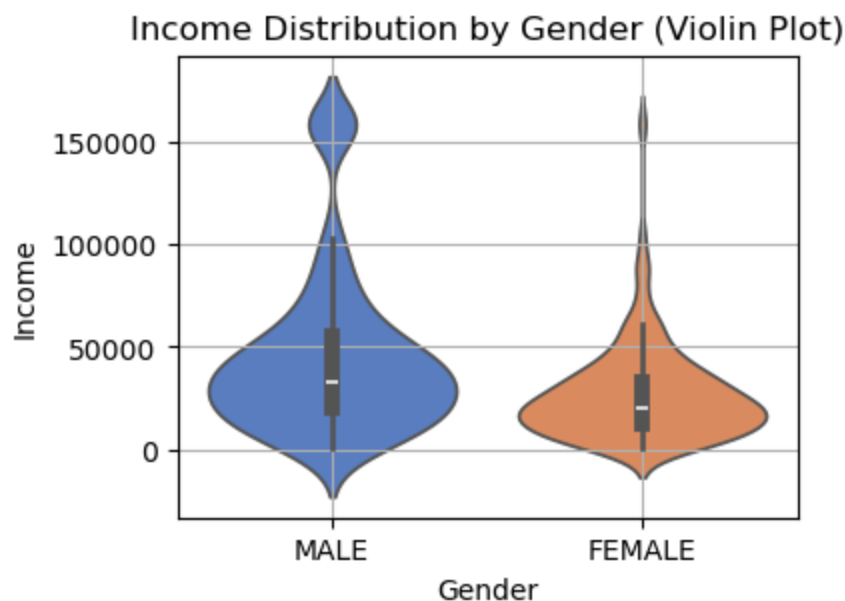
```
In [12]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import warnings
plt.figure(figsize=(8, 6))
sns.regplot(x='age', y='income', data=gss, scatter_kws={'alpha':0.5}, line_kws={'co
```

```
plt.title("Income vs Age (Regression Line)")
plt.xlabel("Age")
plt.ylabel("Income")
plt.show()
```



Violin Plot

```
In [13]: plt.figure(figsize=(4, 3))
sns.violinplot(x='sex', y='income', data=gss, palette='muted')
plt.title("Income Distribution by Gender (Violin Plot)")
plt.xlabel("Gender")
plt.ylabel("Income")
plt.grid(True)
plt.show()
```



Central Tendency

- **Higher male median:** White indicator shows higher male median.
- **Lower female median:** Female median sits noticeably lower.

Distribution and Density

- **Lower income concentration:** Both groups peak at lower incomes.
- **Lower female peak:** Female peak occurs at lower value.
- **Wider male spread:** Male distribution shows greater overall variance.
- **Larger male IQR:** Central dark box is taller for males.

High-Income Patterns

- **Male bimodal peak:** Separate male group exists near 150,000.
- **Female outlier tail:** Long, thin tail reaches 150,000.
- **Scarce female high-earners:** Extremely few females reach top brackets.

```
In [14]: # Create a new figure with specified dimensions (4x3 inches)
plt.figure(figsize=(4, 3))

# Create a violin plot showing income distribution by gender
# Violin plots combine box plots with kernel density estimation to show data distributions
sns.violinplot(x='sex', y='income', data=gss, palette='muted')

# Add descriptive title to the plot
```

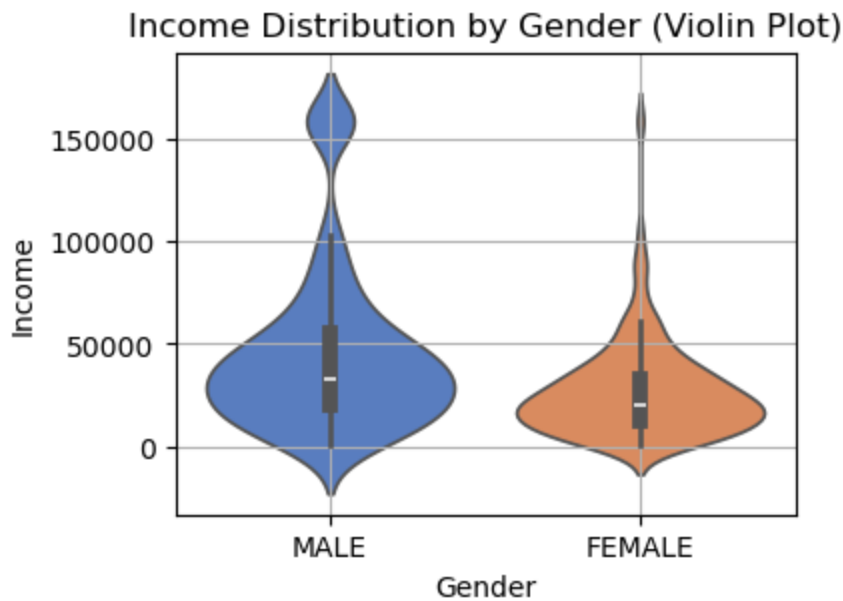
```
plt.title("Income Distribution by Gender (Violin Plot)")

# Set x-axis Label
plt.xlabel("Gender")

# Set y-axis Label
plt.ylabel("Income")

# Add grid lines for better readability
plt.grid(True)

# Display the completed plot
plt.show()
```



Key Insights:

Distribution Shape:

- Shows the full probability density of income for each gender
- Reveals whether income distributions are normal, skewed, or have multiple peaks

Gender Income Comparison:

- Width of each violin indicates the density of data points at different income levels
- Allows comparison of income distribution patterns between male and female groups

Statistical Patterns:

- **Median differences** - horizontal lines inside violins show median income by gender
- **Spread/variability** - wider sections indicate more people at those income levels
- **Outliers and extremes** - thin tails reveal high/low income earners

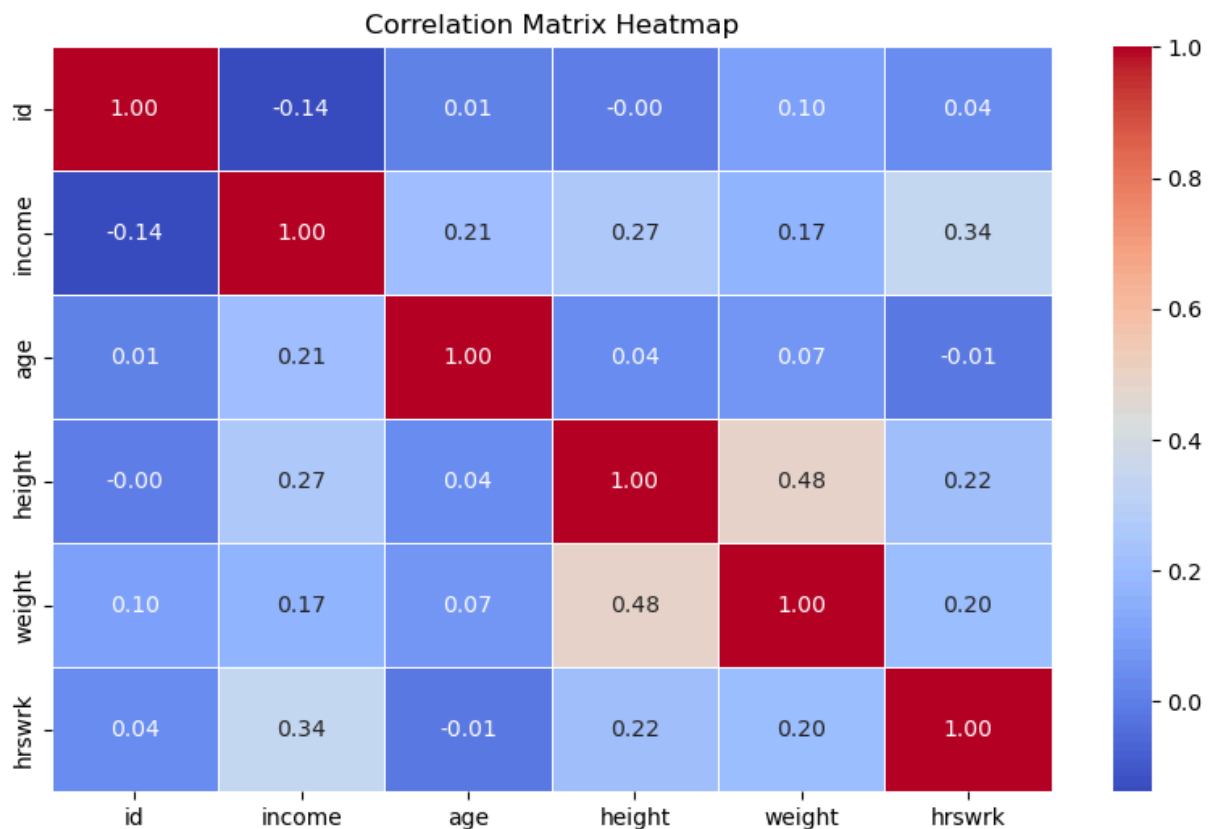
Potential Findings:

- Income inequality between genders
- Whether one gender has more concentrated income ranges
- Presence of income gaps or clustering at certain salary levels

Heatmap (Correlation Matrix)

In [15]: `#gss.info()`

In [16]: `plt.figure(figsize=(10, 6))
corr_matrix = gss.select_dtypes(include=[np.number]).corr()
sns.heatmap(corr_matrix, annot=True, cmap='coolwarm', fmt=".2f", linewidths=0.5)
plt.title("Correlation Matrix Heatmap")
plt.show()`



Expected Output:

- A 10x6 inch correlation matrix heatmap showing relationships between all numerical variables in the `gss` dataset
- Each cell displays the correlation coefficient (formatted to 2 decimal places)
- Color coding using the 'coolwarm' palette where:
 - **Red/warm colors** indicate strong positive correlations (closer to +1)
 - **Blue/cool colors** indicate strong negative correlations (closer to -1)
 - **White/neutral colors** indicate weak or no correlation (closer to 0)
- Thin grid lines (0.5 width) separate each correlation cell

- The diagonal will show perfect correlation (1.00) since each variable correlates perfectly with itself

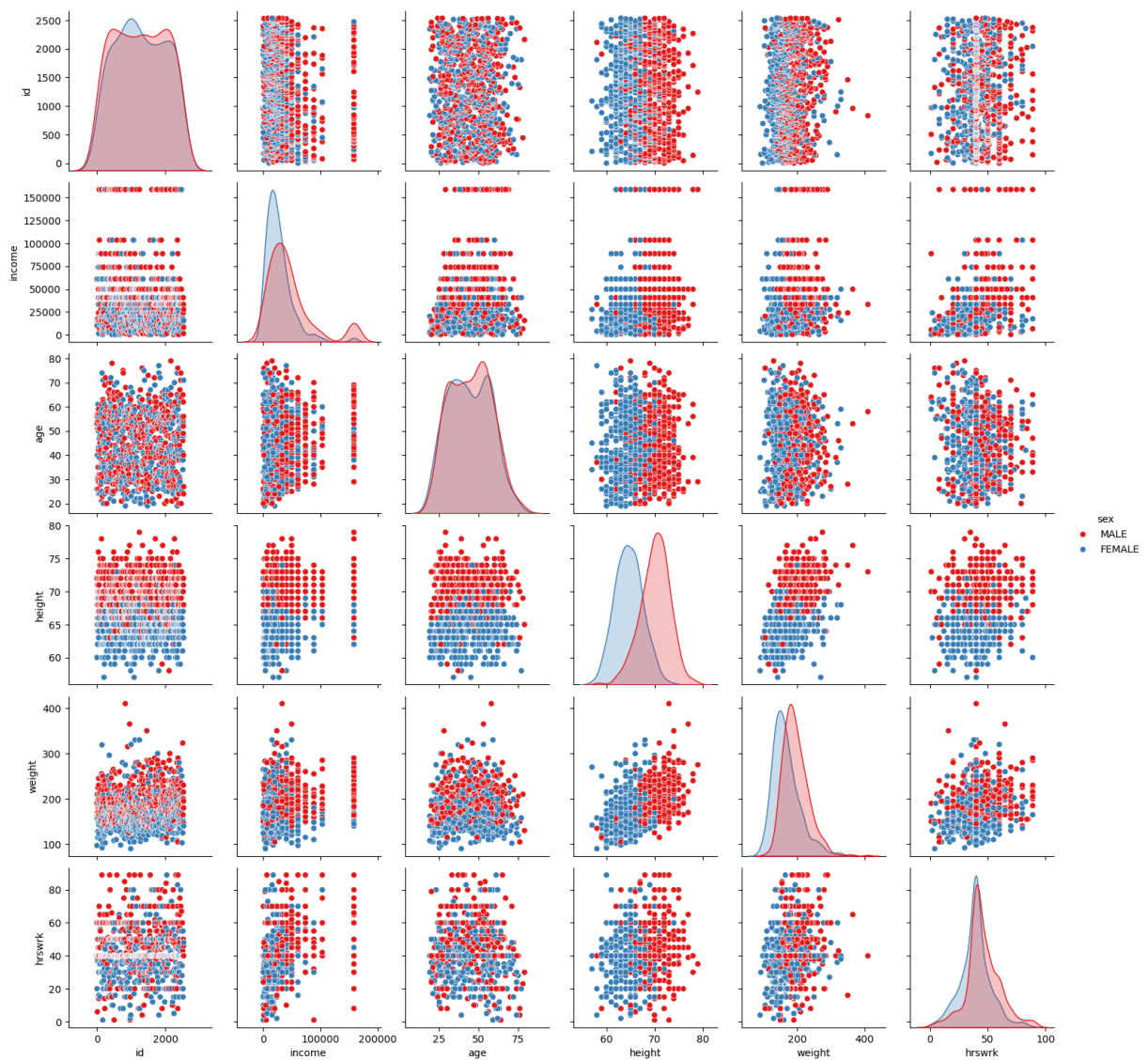
What to look for in the results:

- **Strong correlations** ($|r| > 0.7$): Variables that move together strongly
- **Moderate correlations** ($0.3 < |r| < 0.7$): Variables with meaningful relationships
- **Weak correlations** ($|r| < 0.3$): Variables with little linear relationship
- **Unexpected patterns**: Surprising relationships that warrant further investigation

This visualization helps identify multicollinearity issues, potential feature relationships, and guides feature selection for modeling.

Pair Plot

```
In [17]: sns.pairplot(gss, hue='sex', diag_kind='kde', palette='Set1')
plt.show()
```





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Thank you